

INDUSTRIAL HYDRAULIC POWER UNIT (HPU)

FIELD SERVICE and MAINTENANCE MANUAL

Document Reference: MNL-HPU-2026-H540

Equipment Class: Hydraulic Power Unit with Pump, Valves, and Actuator

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SECTION 1: SYSTEM OVERVIEW

The hydraulic power unit (HPU) supplies pressurized hydraulic fluid to drive actuators such as cylinders and hydraulic motors. The major components a field technician services are: the hydraulic pump, the electric pump motor, the reservoir (fluid tank), the pressure relief valve, the directional control valve, the hydraulic cylinder (actuator), the hydraulic hoses and fittings, and the fluid filter. System pressure is monitored on a gauge and regulated by the relief valve.

Safety: Hydraulic fluid is under HIGH PRESSURE. Always shut down the pump and relieve system pressure to zero on the gauge before opening any fitting, hose, or valve. Never search for a leak with bare hands — pressurized fluid can inject through skin. Wear eye protection.

SECTION 2: HYDRAULIC PUMP

Component identity: The hydraulic pump is the unit mounted to the electric motor that draws fluid from the reservoir and delivers it under pressure to the system.

Failure mode 2.1 — Pump produces low or no pressure

Symptoms: Gauge reads below normal; actuators move slowly or not at all.

Cause: Low fluid level, clogged suction strainer, worn pump, or relief valve stuck open.

Corrective steps:

1. Shut down and relieve pressure to zero. Lock out power.
2. Check reservoir fluid level; top up to the sight-glass mark with the specified oil.
3. Inspect and clean the suction strainer at the pump inlet.
4. Verify the pressure relief valve is not stuck open (see Section 5).
5. If pressure is still low with full fluid and a clean strainer, the pump is worn and must be rebuilt or replaced.

Failure mode 2.2 — Pump is noisy (cavitation or aeration)

Symptoms: Loud whining, knocking, or rattling from the pump; foamy fluid.

Cause: Restricted suction (cavitation) or air entering the suction side (aeration).

Corrective steps:

1. Shut down and relieve pressure. Lock out power.
2. Check fluid level; low fluid lets the pump draw air.
3. Clean the suction strainer and confirm the suction line is not collapsed or kinked.
4. Tighten all suction-side fittings to eliminate air ingress.
5. Bleed air from the system after restarting.

Failure mode 2.3 — Pump overheating

Symptoms: Pump and fluid hot to the touch; thinning fluid; reduced performance.

Cause: Continuous relief at high pressure, low fluid, or a failing pump.

Corrective steps:

1. Shut down and let the unit cool. Lock out power.
2. Check fluid level and top up; low fluid reduces heat dissipation.
3. Verify the system is not running against the relief valve continuously (a stuck valve or blocked line forces all flow over relief, generating heat).
4. Inspect the cooler if fitted; clean its fins.

SECTION 3: ELECTRIC PUMP MOTOR

Component identity: The electric motor drives the hydraulic pump; it has a cooling fan cover and a terminal box.

Failure mode 3.1 — Motor will not start

Symptoms: Motor hums or is silent; pump does not run.

Cause: Lost supply phase, failed contactor, tripped overload, or a seized pump.

Corrective steps:

1. Lock out power. Try to turn the pump shaft by hand; if seized, the pump is the fault (see Section 2).
2. Check all supply phases at the terminal box; a missing phase causes humming.
3. Reset and inspect the overload relay and contactor.

Failure mode 3.2 — Motor overheating

Symptoms: Motor housing very hot; thermal trip; burning smell.

Cause: Overload (high system pressure), blocked cooling fan, or low voltage.

Corrective steps:

1. Stop the motor and let it cool. Lock out power.
2. Clear debris from the fan cover and cooling fins.
3. Measure supply voltage at the terminal box against the nameplate.
4. If the system runs at high pressure continuously, reduce duty cycle or check for a stuck relief valve forcing the motor to work harder.

SECTION 4: RESERVOIR AND FLUID

Component identity: The reservoir is the tank that holds the hydraulic fluid; it has a sight glass, a fill/breather cap, and a drain plug.

Failure mode 4.1 — Low fluid level

Symptoms: Sight glass below minimum; pump noise; erratic actuator motion.

Cause: External leak, internal consumption, or insufficient initial fill.

Corrective steps:

1. Shut down and relieve pressure. Inspect for external leaks (hoses, fittings, seals).
2. Repair any leak found before refilling.
3. Top up with the specified hydraulic oil to the sight-glass mark.

Failure mode 4.2 — Contaminated or degraded fluid

Symptoms: Cloudy, dark, or foamy fluid; particles in the sight glass; varnish smell.

Cause: Water ingress, particulate contamination, or fluid past service life.

Corrective steps:

1. Shut down and relieve pressure.
2. Drain the reservoir fully and replace the fluid filter.
3. Refill with fresh specified hydraulic oil.
4. Investigate the contamination source (failed breather, seal, or cooler).

SECTION 5: PRESSURE RELIEF VALVE

Component identity: The pressure relief valve protects the system by venting excess fluid back to the reservoir above a set pressure.

Failure mode 5.1 — Relief valve stuck open (low system pressure)

Symptoms: System pressure never reaches setpoint; fluid heats up.

Cause: Debris under the valve seat, worn poppet, or weak spring.

Corrective steps:

1. Shut down and relieve pressure. Lock out power.
2. Remove and inspect the relief valve poppet and seat for debris or wear.
3. Clean or replace the valve; verify the spring and adjustment setting.
4. Re-set the relief pressure to the specified value and confirm on the gauge.

Failure mode 5.2 — Relief valve stuck closed (over-pressure)

Symptoms: System pressure rises above setpoint; hoses bulge; safety risk.

Cause: Seized poppet or incorrect adjustment.

Corrective steps:

1. STOP IMMEDIATELY and relieve pressure — over-pressure is dangerous.
2. Replace the relief valve; do not attempt to run until corrected.
3. Verify the new valve relieves at the specified pressure on the gauge.

SECTION 6: DIRECTIONAL CONTROL VALVE

Component identity: The directional control valve (often solenoid-operated) routes fluid to extend or retract the actuator.

Failure mode 6.1 — Actuator does not move or moves the wrong way

Symptoms: Cylinder will not extend/retract, or moves opposite to the command.

Cause: Failed solenoid coil, stuck spool, or reversed connections.

Corrective steps:

1. Shut down and relieve pressure. Lock out power.
2. Check the solenoid coil for continuity; replace if open.
3. Manually actuate the valve override (if fitted) to confirm spool movement.
4. Free or replace a stuck spool; verify port connections are correct.

SECTION 7: HYDRAULIC CYLINDER (ACTUATOR)

Component identity: The hydraulic cylinder is the actuator that converts fluid pressure into linear motion via a rod and piston.

Failure mode 7.1 — Cylinder drifts or creeps under load

Symptoms: The rod slowly moves on its own when it should hold position.

Cause: Worn piston seals (internal bypass) or a leaking control valve.

Corrective steps:

1. Shut down and relieve pressure.
2. Isolate whether the drift is the cylinder (internal seal bypass) or the valve.
3. Reseal or replace the cylinder piston seals if internal bypass is confirmed.

Failure mode 7.2 — External fluid leak at the cylinder rod

Symptoms: Fluid weeping from the rod gland/seal during operation.

Cause: Worn rod seal or scored rod.

Corrective steps:

1. Shut down and relieve pressure. Lock out power.
2. Replace the rod seal kit.
3. Inspect the rod for scoring; a scored rod will quickly destroy a new seal and must be repaired or replaced.

SECTION 8: HOSES AND FITTINGS

Component identity: Hoses and fittings carry pressurized fluid between components.

Failure mode 8.1 — Hose or fitting leak

Symptoms: Fluid seeping or spraying from a hose, coupling, or fitting.

Cause: Loose fitting, worn hose, or failed seal/O-ring.

Corrective steps:

1. Shut down and relieve pressure to ZERO before touching any fitting.
2. Tighten the fitting to spec if it is the source.
3. Replace a damaged or perished hose and any failed O-rings.
4. Never use hands to locate a pressurized leak — use cardboard.

SECTION 9: ROUTINE MAINTENANCE SCHEDULE

Daily: Check fluid level in the sight glass; check system pressure reaches setpoint; look for visible leaks.

Weekly: Inspect hoses and fittings for wear or seepage; feel the pump and motor for excess heat after a run.

Monthly: Check fluid condition (color, clarity); inspect the filter indicator; verify the relief valve setting on the gauge.

Quarterly: Change the fluid filter; sample fluid for contamination; check cylinder rod seals for weeping.

End of Manual — MNL-HPU-2026-H540